

1(a) The probability distribution of a random variable X is modelled as follows.

$$P(X = x) = \begin{cases} \frac{k}{x} & x = 1, 2, 3, 4, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

Show that $k = \frac{12}{25}$

[2]

(b) Show in a table the values of X and their probabilities.

[1]

(c) The values of three independent observations of X are denoted by X_1 , X_2 and X_3 .

Find $P(X_1 > X_2 + X_3)$.

[3]

- (d) In a game, a player notes the values of successive independent observations of X and keeps a running total. The aim of the game is to reach a total of exactly 7.

Determine the probability that a total of exactly 7 is first reached on the 5th observation.

[5]

2(a) The probability distribution of a random variable X is shown in the table, where p is a constant.

x	0	1	2	3
$P(X=x)$	$\frac{1}{12}$	$\frac{1}{4}$	p	$3p$

Two values of X are chosen at random. Determine the probability that their product is greater than their sum. [5]

- (b) A random variable Y takes n values, each of which is equally likely. Two values, Y_1 and Y_2 , of Y are chosen at random.

It is given that $P(Y_1 = Y_2) = 0.02$.

Find $P(Y_1 > Y_2)$.

[2]

3(a) In this question you must show detailed reasoning.

The random variable X has probability distribution defined as follows.

$$P(X = x) = \begin{cases} \frac{15}{64} \times \frac{2^x}{x!} & x = 2, 3, 4, 5, \\ 0 & \text{otherwise.} \end{cases}$$

Show that $P(X = 2) = \frac{15}{32}$. [1]

(b) The values of three independent observations of X are denoted by X_1 , X_2 and X_3 .

Given that $X_1 + X_2 + X_3 = 9$, determine the probability that at least one of these three values is equal to 2. [6]

(c) Freda chooses values of X at random until she has obtained $X = 2$ exactly three times. She then stops.

Determine the probability that she chooses exactly 10 values of X .

[3]

4(a) In this question you must show detailed reasoning.

A biased four-sided spinner has edges numbered 1, 2, 3, 4. When the spinner is spun, the probability that it will land on the edge numbered X is given by

$$P(X = x) = \begin{cases} \frac{1}{2} - \frac{1}{10}x & x = 1, 2, 3, 4, \\ 0 & \text{otherwise.} \end{cases}$$

Draw a table showing the probability distribution of X .

[1]

(b) The spinner is spun three times and the value of X is noted each time.

Find the probability that the third value of X is greater than the sum of the first two values of X .

[3]

5(a) A random variable X has probability distribution defined as follows.

$$P(X = x) = \begin{cases} kx & x = 1, 2, 3, 4, 5, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

Show that $P(X = 3) = 0.2$.

[3]

(b) Show in a table the values of X and their probabilities.

[2]

(c) Two independent values of X are chosen, and their total T is found.

(i) Find $P(T = 7)$.

[3]

(ii) Given that $T = 7$, determine the probability that one of the values of X is 2.

[4]

6 Bag A contains 3 black discs and 2 white discs only. Initially Bag B is empty. Discs are removed at random from bag A, and are placed in bag B, one at a time, until all 5 discs are in bag B.

(a) Write down the probability that the last disc that is placed in bag B is black. [1]

(b) Find the probability that the first disc and the last disc that are placed in bag B are both black. [2]

(c) Find the probability that, starting from when the first disc is placed in bag B, the number of black discs in bag B is always greater than the number of white discs in bag B. [4]

- 7 Joanne has five cards, numbered 1, 1, 1, 2, 2. She picks two cards at random, without replacement. The variable X denotes the sum of the numbers on the two cards.

(a) Show that $P(X = 3) = \frac{3}{5}$. [2]

The table shows the probability distribution of X .

x	2	3	4
$P(X = x)$	$\frac{3}{10}$	$\frac{3}{5}$	$\frac{1}{10}$

Joanne replaces the two cards. Now Liam picks two cards at random from the five cards, without replacement. The variable Y denotes the sum of the numbers on the two cards that Liam picks.

(b) Find $P(X = Y)$. [2]

- 8 A random variable X has probability distribution given by

$$P(X = x) = \frac{1}{860}(1 + x) \quad \text{for } x = 1, 2, 3, \dots, 40.$$

(a) Find $P(X > 39)$. [2]

(b) Given that x is even, determine $P(X < 10)$. [6]

9 The probability distribution of a random variable X is given in the table.

x	1	2	3
$P(X = x)$	0.6	0.3	0.1

Two values of X are chosen at random. Find the probability that the second value is greater than the first.

[3]

10(a) Anika and Beth are playing a game which consists of several points.

- The probability that Anika will win any point is 0.7.
- The probability that Beth will win any point is 0.3.
- The outcome of each point is independent of the outcome of every other point.

The first player to win two points wins the game.

Write down the probability that the game consists of more than three points.

[1]

(b) Draw a tree diagram to show all the possibilities for the game.

[3]

(c) Determine the probability that Beth wins the game.

[3]

(d) Determine the probability that the game consists of exactly three points.

[2]

(e) Given that Beth wins the game, determine the probability that the game consists of exactly three points.

[4]

11 Mr Jones has 3 tins of beans and 2 tins of pears. His daughter has removed the labels for a school project, and the tins are identical in appearance. Mr Jones opens tins in turn until he has opened at least 1 tin of beans and at least 1 tin of pears. He does not open any remaining tins.

(a) Draw a tree diagram to illustrate this situation, labelling each branch with its associated probability. [3]

(b) Find the probability that Mr Jones opens exactly 3 tins. [3]

(c) It is given that the last tin Mr Jones opens is a tin of pears. Find the probability that he opens exactly 3 tins. [5]

12(a) Andy and Bev are playing a game.

- The game consists of three points.
- On each point, $P(\text{Andy wins}) = 0.4$ and $P(\text{Bev wins}) = 0.6$.
- If one player wins two consecutive points, then they win the game, otherwise neither player wins.

Determine the probability of the following events.

(i) Andy wins the game.

[2]

(ii) Neither player wins the game.

[3]

(b) Andy and Bev now decide to play a match which consists of a series of games.

- In each game, if a player wins the game then they win the match.
- If neither player wins the game then the players play another game.

Determine the probability that Andy wins the match.

[3]

- 13 Rob has two six-sided dice, each with sides numbered 1, 2, 3, 4, 5, 6.
One dice is fair. The other dice is biased, with probabilities as shown in the table.

Biased die						
y	1	2	3	4	5	6
$P(Y = y)$	0.3	0.25	0.2	0.14	0.1	0.01

Rob throws each dice once and notes the two scores, X on the fair dice and Y on the biased dice. He then calculates the value of the variable S which is defined as follows.

- If $X \leq 3$, then $S = X + 2Y$.
- If $X > 3$, then $S = X + Y$.

- (a) (i) Draw up a sample space diagram showing all the possible outcomes and the corresponding values of S . [2]

- (ii) On your diagram, circle the four cells where the value $S = 10$ occurs.

[1]

(b) Explain the mistake in the following calculation.

[1]

$$P(S = 10) = \frac{\text{Number of outcomes giving } S = 10}{\text{Total number of outcomes}} = \frac{4}{36} = \frac{1}{9}.$$

(c) Find the correct value of $P(S = 10)$.

[2]

(d) Given that $S = 10$, find the probability that the score on one of the dice is 4.

[3]

(e) The events “ $X = 1$ or 2 ” and “ $S = n$ ” are mutually exclusive. Given that $P(S = n) \neq 0$, find the value of n . [1]

END OF QUESTION PAPER

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
1	a	$k(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}) = 1$ $k \times \frac{25}{12} = 1$ or eg $\frac{25}{3}k = 4$ or $25k = 12$. hence $k = \frac{12}{25}$ AG Alternative method or verification: $\frac{12}{25} + \frac{6}{25} + \frac{4}{25} + \frac{3}{25} = 1$	M1 A1 Alternative method M1 A1 [2]	Correct equation involving multiple of k Must see previous line and answer
	b	1 2 3 4 $\frac{12}{25} \frac{6}{25} \frac{4}{25} \frac{3}{25}$	B1 [1]	or equivalent exact values
	c	(3, 1, 1) (4, 1, 1) (4, 2, 1) (4, 1, 2) $\frac{4}{25} \times (\frac{12}{25})^2 + \frac{3}{25} \times (\frac{12}{25})^2 + \frac{3}{25} \times \frac{6}{25} \times \frac{12}{25}$ $+ \frac{3}{25} \times \frac{12}{25} \times \frac{6}{25}$ $= \frac{288}{3125}$ or 0.09216	M1 M1 A1 [3]	At least three of these seen or implied. No extras or repeats. At least two correct terms, no incorrect coefficients; ft their table. Allow in terms of k Allow 0.0922 (3 sf) <u>Examiner's Comments</u>  Misconception A common misconception was to interpret $P(X^1 > X^2 + X^3)$ to mean that it is required to find three values of X such that the probability of the 1st value is greater than the sum of the probabilities of the 2nd and 3rd values.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	d		(1, 1, 1, 1, 3) (1, 1, 1, 2, 2) $\left(\frac{12}{25}\right)^4 \times \frac{4}{25} \times 5 + \left(\frac{12}{25}\right)^3 \times \left(\frac{6}{25}\right)^2 \times {}^5C_2$ Alternative method for M1A1 $\text{or } \left(\frac{12}{25}\right)^4 \times \frac{4}{25} \times (4+1) + \left(\frac{12}{25}\right)^3 \times \left(\frac{6}{25}\right)^2 \times ({}^4C_2+4)$ $= \frac{41472}{390625} \text{ or } 0.10616832$	B1 B1 M1 A1 Alternative method for M1A1 M1 A1 A1 [5]	B1B1 for both sets in any order, without extras. Both soi. B1 for both sets in any order, with extras. $\left(\frac{12}{25}\right)^4 \times \frac{4}{25}$ or $\left(\frac{12}{25}\right)^3 \times \left(\frac{6}{25}\right)^2$ oe seen. Ignore coeffs. ft their table For either $\left(\frac{12}{25}\right)^4 \times \frac{4}{25} \times 5$ or $\left(\frac{12}{25}\right)^3 \times \left(\frac{6}{25}\right)^2 \times {}^5C_2$ oe ft their table Alternative method for M1A1 oe For either $\left(\frac{12}{25}\right)^3 \times \left(\frac{6}{25}\right)^2 \times ({}^4C_2+4)$ oe Allow 0.106 (3 sf)
			Total	11	
2	a		$\frac{1}{p} + \frac{1}{4} + p + 3p = 1$ $p = \frac{1}{6}$ (2,3) (3,2) (3,3) $\frac{1}{5} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{6} + \frac{1}{2} \times \frac{1}{2}$ $\frac{5}{12}$	M1 A1 M1 M1 A1 [5]	soi. Allow M1 for any two correct pairs and no incorrect pairs or for all three correct pairs and at most one incorrect pair All correct and added, ft their p.
	b		$\frac{1}{2}(1 - 0.02)$ $= 0.49$	M1 A1 [2]	
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3	a		DR $\frac{15}{64} \times \frac{2^2}{2!}$ oe eg $(= \frac{15}{32} \text{ AG})$ $\frac{15}{64} \times \frac{4}{2}$	B1 (AO1.1) [1]	Must see this expression and result

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
b		DR 2, 2, 5 2, 3, 4 3, 3, 3 $P(X_1 + X_2 + X_3 = 9) =$ $3 \times \left(\frac{15}{32}\right)^2 \times \frac{5}{80} + 6 \times \frac{15}{32} \times \frac{5}{16} \times \frac{5}{32} + \left(\frac{5}{16}\right)^3$ 0.0412 + 0.1373 + 0.0305 $3 \times \frac{225}{16384} + 6 \times \frac{375}{16384} + \frac{125}{4096}$ $\frac{675}{16384} + \frac{1125}{8192} + \frac{125}{4096} \quad (= 0.209045)$ $P(X_1 + X_2 + X_3 = 9 \text{ and at least 1 } X \text{ value} = 2)$ $= 3 \times \left(\frac{15}{32}\right)^2 \times \frac{5}{80} + 6 \times \frac{15}{32} \times \frac{5}{16} \times \frac{5}{32} \quad (= 0.178528)$ $\frac{'0.178528'}{'0.209045'}$ = 0.854 (3 sf) or $\frac{117}{137}$ $P(X_1 + X_2 + X_3 = 9 \text{ and no } X \text{ value} = 2)$ $= \left(\frac{5}{16}\right)^3 \quad (= 0.030518 \text{ or } \frac{125}{4096})$ $1 - \frac{'0.030518'}{'0.209045'}$ = 0.854 (3 sf) or $\frac{117}{137}$	M1 (AO3.1a) M1 (AO2.1) M1 (AO2.1) M1 (AO1.1) M1 (AO2.1) A1 (AO2.2a) M1 M1 A1 [6]	Any two seen, with no more than 2 extra different combinations. eg 0, 4, 5 and 0, 5, 4 count as <u>one</u> extra M2: ≥ 1 correct product actually seen & all three products correct M1: 1 correct product seen or all correct except omission of, or incorrect, multiple(s) or all three results or total correct, but without working Allow M1 for 1 correct product or omit, or incorrect, multiple(s) or ft their probabilities from their previous calculation $\div$ their attempted probs of correct events ft their $P(3, 3, 3)$ $\div$ their attempted probabilities of correct events & subtract from 1 NB $1 - \left(\frac{5}{16}\right)^3$ alone scores M1 <u>Examiner's Comments</u> Most candidates were unable to find their way through the various calculations needed. Some obtained a correct product of three probabilities (e. g. for 2, 2, 5) but omitted to multiply by 3. Others omitted the case 3, 3, 3 altogether.
c		P(two 2's in nine vales of X) or 0.094466 or ${}^9C_2 \times \left(1 - \frac{15}{32}\right)^7 \times \left(\frac{15}{32}\right)^2$ $P(\text{two 2's in nine vales of } X) \times P(X = 2)$	M1 (AO3.1a) M1 (AO2.1) A1 (AO1.1)	soi eg by 9C_2 seen soi

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			or $0.094466 \times \frac{15}{32}$ or ${}^9C_2 \times (1 - \frac{15}{32})^7 \times (\frac{15}{32})^3$ 0.0443 (3 sf)	[3]	NB $(\frac{17}{32})^7 \times (\frac{15}{32})^3$ scores 0, unless multiplied by 9C_2 <u>Examiner's Comments</u> Most candidates did not recognise the need to consider the first 9 values separately before considering the 10th. Some candidates seemed unfamiliar with the binomial distribution.  Misconception In answering Question 7 (about vectors) some candidates appeared not to appreciate the different significances of a, $\mathbf{a}$ and A. Some candidates appeared not to understand the description "cartesian equation" and attempted to find some sort of vector equation of the circle in question 7(d). Many candidates did not appreciate the need for the modulus sign, and how to use it, after integrating $\frac{1}{P-100}$, in question 8. Many candidates used '$=$' instead of '$\leq$' in the hypothesis test, as mentioned in Question 10. In answering Question 13(b), most candidates did not recognise that an infinite series was involved.
			Total	10	

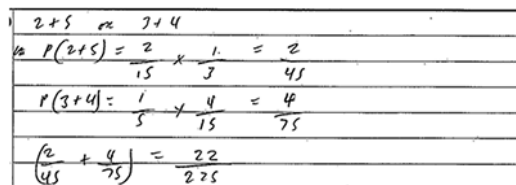
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4	a		$\frac{1}{5} \quad \frac{2}{10} \quad \frac{3}{5} \quad \frac{4}{10} \quad \text{oe}$	B1(AO1.1) [1]	B1
	b		DR 1,1,3; 1,1,4; 1,2,4; 2,1,4 $\left(\frac{2}{5}\right)^2 \times \frac{1}{5} + \left(\frac{2}{5}\right)^2 \times \frac{1}{10} + \frac{2}{5} \times \frac{3}{10} \times \frac{1}{10} + \frac{3}{10} \times \frac{2}{5} \times \frac{1}{10}$ $= \frac{9}{125} \text{ or } 0.072$	B1(AO3.1a) M1(AO2.1) A1(AO1.1) [3]	B1 for any three of these soi, eg on tree diagram Any two correct products added: ft their table <u>Examiner's Comments</u> Many candidates did not identify all four cases. Some only found the products of two, rather than three, probabilities.
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
5	a		$k(1 + 2 + 3 + 4 + 5) (= 1)$ $k = \frac{1}{15}$ $P(X = 3) = 3 \times \frac{1}{15}$ or $\frac{3}{15}$ (= 0.2 AG)	M1(AO3.3) A1(AO1.1) A1(AO2.1) [3]	Allow $15k (= 1)$ May be implied Must see $3 \times \frac{1}{15}$ or $\frac{3}{15}$ and answer 0.2	<u>Examiner's Comments</u> Most candidates answered this question correctly. Many, however, (perhaps misunderstanding the phrase "where k is a constant") assumed that all five probabilities were equal, and therefore each probability was 0.2
	b		1 2 3 4 5 $\frac{1}{15}$ $\frac{2}{15}$ $\frac{3}{15}$ $\frac{4}{15}$ $\frac{5}{15}$ oe 0.07, 0.13, 0.2, 0.27, 0.33	M1(AO1.2) A1(AO1.1) [2]	M1 for ≥ 3 probs correct, ft their k cao. Allow decimals (2 dp) SC: Table with all five probs = 0.2: M1	Allow $X = 0$ or $X = 6$ or $X = 6+$ if prob shown as 0 <u>Examiner's Comments</u> Almost all candidates answered this correctly, or incorrectly but consistently with their method for part (a).
	c	i	Both parts. Allow mixture of methods $\frac{3}{15} \times \frac{4}{15} + \frac{2}{15} \times \frac{5}{15}$ oe $\times 2$ $\frac{44}{225}$ or 0.196 (3 sf)	M1(AO3.4) M1(AO3.1a) A1(AO1.1)	Correct products added, ft their table $2 \times (\text{Sum of two products of probs})$ cao Special cases 2-way table Count 4 pairs M1	All probs = 0.2 $0.2^2 + 0.2^2$ M1

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
				[3]	But if (b) correct: M0 ÷ 2 5 M1 = 0.16 A0	But if (b) correct: M0 2×(0.2 ² +0.2 ²) Allow without 2× M1 = 0.16 A0
<u>Examiner's Comments</u>						
In both these parts, some candidates either assumed that the probabilities for all the values of X were equal, or (which is equivalent) drew up a two-way table of all the possible outcomes and counted up the relevant ones. These were able to gain only a few marks. Some candidates used one of these incorrect methods, despite having answered parts (a) and (b) correctly.						
Many candidates recognised that for T = 7 could be found by considering 2+5 and 3+4, but forgot to consider 5+2 and 4+3.						
Exemplar 1						
						
This candidate has shown clear working, but forgotten to multiply by two in order to consider all the ways to score T = 7 .						

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance																				
		ii	$P(\text{one value is 2 \& } T = 7) = 2 \times \frac{2}{15} \times \frac{5}{15}$ $= \frac{4}{45}$ $\frac{P(\text{one value is 2 \& } T = 7)}{P(T = 7)} \quad \left(\frac{\frac{4}{45}}{\frac{44}{225}} \text{ or } = \frac{0.0889}{0.196} \right)$ $= \frac{5}{11} \text{ or } 0.455 \text{ (3 sf)}$	M1(AO3.4) A1f(AO1.1) M1(AO2.1) A1(AO1.1) [4]	Allow without "2×", ft their table ft their table (except if all probs = 0.2) Allow any probability Their (c)(i) or their $P(T = 7)$ cao NB not 0.454 Eg: If (i) $\frac{22}{225} \div \frac{2}{45} \div \frac{22}{225} = \frac{5}{11}$ M1A0M1A0 Count 2 pairs <table><tr><td></td><td>M1</td><td></td><td></td></tr><tr><td></td><td>A0</td><td>2×0.2^2 M1</td><td></td></tr><tr><td></td><td></td><td>$= 0.08$</td><td>A0</td></tr><tr><td>$\div 4$</td><td>M1</td><td>$\frac{0.08}{0.16}$ M1</td><td></td></tr><tr><td>$= 0.5$</td><td>A0</td><td>$= 0.5$</td><td>A0</td></tr></table> Examiner's Comments Most candidates attempted to use the conditional probability formula and, correctly, placed their answer to part (c)(i) in the denominator of a fraction. But many were unable to identify the pairs of values that gave rise to "T = 7 AND one of the values is 2".		M1				A0	2×0.2^2 M1				$= 0.08$	A0	$\div 4$	M1	$\frac{0.08}{0.16}$ M1		$= 0.5$	A0	$= 0.5$	A0
	M1																								
	A0	2×0.2^2 M1																							
		$= 0.08$	A0																						
$\div 4$	M1	$\frac{0.08}{0.16}$ M1																							
$= 0.5$	A0	$= 0.5$	A0																						
			Total	12																					

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6		a	$\frac{3}{5}$	B1 (AO1.1) [1]	
		b	$\frac{3}{5} \times \frac{2}{4}$ $= \frac{3}{10}$	M1 (AO1.1a) A1 (AO1.1) [2]	or $\frac{3}{5} \times (\frac{2}{4} \times \frac{2}{3} \times \frac{1}{2} \times 3)$ or $\frac{3}{5} \times \frac{{}^2C_2 \times {}^2C_1}{{}^4C_3}$
		c	BBB, BBWB $\frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} + \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} \times \frac{1}{2}$ $= \frac{1}{5}$ oe	M1 (AO3.1b) M1 (AO2.1) M1 (AO1.1a) A1 (AO1.1) [4]	With no extras M1 for each correct product of probs or $\frac{2}{\frac{5!}{3!2!}}$
			Total	7	
7		a	$\frac{3}{5} \times \frac{1}{2}$ or $\frac{2}{5} \times \frac{3}{4}$ $\frac{3}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{3}{4}$ $(= \frac{3}{5} \text{ AG})$	M1(AO1.1) A1(AO1.1) [2]	or $\frac{3}{5} \times \frac{1}{2} \times 2$ or $\frac{2}{5} \times \frac{3}{4} \times 2$ Must see this step
		b	$\left(\frac{3}{5}\right)^2 + \left(\frac{3}{10}\right)^2 + \left(\frac{1}{10}\right)^2$ $= \frac{23}{50}$ or 0.46	M1(AO1.1a) A1(AO1.1) [2]	
			Total	4	

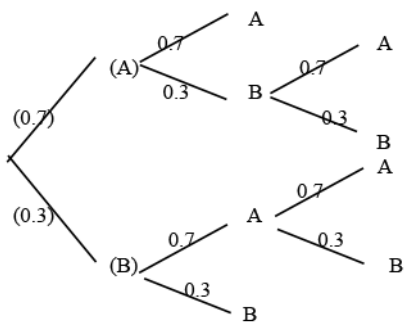
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
8		a	$P(X > 39) = P(X = 40) = \frac{1}{860}(1 + 40)$ $= \frac{41}{860}$	M1(AO1.1) A1(AO1.1) [2]	Attempt at evaluating $P(X = 40)$	
		b	$P(X \text{ even}) = \frac{1}{860}(20 + (2 + 4 + 6 + \dots + 40)) \text{ oe}$ $= \frac{1}{860}\left(20 + \frac{2+40}{2} \times 20\right)$ $= \frac{22}{43}$ $P(X = 2, 4, 6, 8) = \frac{1}{860}(4 + 2 + 4 + 6 + 8)$ $= \frac{12}{430} \text{ oe}$ $\frac{P(X = 2, 4, 6, 8 \text{ and } X \text{ even})}{P(X \text{ even})} = \frac{P(X = 2, 4, 6, 8)}{P(X \text{ even})}$ $= \frac{12}{430} \div \frac{22}{43} = \frac{3}{55} \text{ oe or } 0.0545 \text{ (3 s.f.)}$	M1(AO3.1a) A1(AO1.1) A1(AO1.1) M1(AO1.1) A1(AO3.2a) B1(AO2.1) [6]	Attempt Σ probabilities of all even values Correct expression Attempt Σ probabilities for $X = 2, 4, 6, 8$ their $P(X = 2, 4, 6, 8)$ their $P(X \text{ even})$ For a clear solution allowing the line of reasoning to be followed, with each component of the conditional probability found clearly	Numerical sums may be evaluated BC throughout
			Total	8		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
9			0.6×0.3 or 0.6×0.1 or 0.3×0.1 $0.6 \times 0.3 + 0.6 \times 0.1 + 0.3 \times 0.1$ oe $= 0.27$	M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) [3]	Any correct product seen, oe Fully correct method	OR M1 $0.6^2 + 0.3^2 + 0.1^2 (= 0.46)$ M1 0.5 × (1-'0.46')
			Total	3		

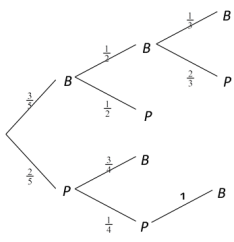
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
10	a		0	B1 [1]	Allow 0%
	b			B1 B1 B1 [3]	Ignore extra branches if no probabilities or $p = 0$ B2: 8 correct branches and probs <u>OR</u> names, no extra branches B2: 7 correct branches, probs and names, no extra branches B1: 8 correct branches without probs & names. No extra branches B1: 6 correct branches, probs and names. Ignore extra branches Ignore products at ends
	c		$0.3 \times 0.3 + 0.7 \times 0.3 \times 0.3 + 0.3 \times 0.7 \times 0.3$ or $0.09 + 0.063 + 0.063$ oe $= \frac{27}{125}$ or 0.216	M1 M1 A1 [3]	All correct M2 ft their diagram Two products correct M1 ft their diagram SC Correct answer with no working: B2
	d		$0.7 \times 0.3 + 0.3 \times 0.7$ or $0.21 + 0.21$ $= \frac{21}{50}$ or 0.42	M1 A1 [2]	or $1 - (0.7^2 + 0.3^2)$ or $1 - (0.49 + 0.09)$ Condone missing brackets or $0.7 \times 0.3 \times 0.7 + 0.7 \times 0.3 \times 0.3 + 0.3 \times 0.7 \times 0.7 + 0.3 \times 0.7 \times 0.3$ oe or $2 \times 0.147 + 2 \times 0.063$ oe Wholly correct method ft their diagram SC Correct answer with no working: B1

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	e		P(B wins and 3 points) $= 0.7 \times 0.3 \times 0.3 + 0.3 \times 0.7 \times 0.3$ or 2×0.063 $(= 0.126 \text{ oe})$ $\frac{P(\text{B wins \& 3 points})}{P(\text{B wins})} = \frac{'0.126'}{'0.216'}$ $= \frac{7}{12}$	M2 M1 A1 [4]	ft their diagram for M-marks soi M1 for one correct product or 0.063 Must attempt $\frac{P(\text{B wins \& 3 points})}{\text{Their (c) NOT (d) or } 0.3 \times 0.3 + 0.7 \times 0.3 \times 0.3 + 0.3 \times 0.7 \times 0.3}$ Allow 0.583 (3 sf) SC no working $\frac{7}{12}$ B2 But minimal working is OK, eg $\frac{2 \times 0.063}{0.216} = \frac{7}{12}$ M2M1A1 Examiner's Comments A common error in the numerator was to use $P(A \cap B) = P(A) \times P(B)$ rather than $P(A \cap B) = P(A) \times P(B/A)$, i.e. in this case the numerator was incorrectly evaluated using 0.216×0.42 instead of $2 \times 0.3^2 \times 0.7$.
			Total	13	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
11		a	Correct structure, Correct labels, b and p Correct probs	B1 (AO 1.1a) B1 (AO 1.1) B1 (AO 1.1) [3]	 allow extra branches with probs = 1
		b	$\frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} + \frac{2}{5} \times \frac{1}{4} (\times 1)$ $= \frac{3}{10}$	M1 (AO 3.1b) M1 (AO 2.1) A1 (AO 1.1) [3]	M1 each correct prod of probs
		c	P(opens 3 and last is pear) $= \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} \quad (= \frac{1}{5})$ P(last is pear) $= \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} + \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} + \frac{2}{5} \times \frac{2}{4} \quad (= \frac{3}{5})$ $\frac{\text{P(opens 3 and last is pear)}}{\text{P(last is pear)}}$ $= \frac{1}{3}$	M1 (AO 1.1) M1 (AO 3.1b) M1 (AO 2.1) M1 (AO 2.4) A1 (AO 2.2a) [5]	correct product or M1 for two correct products added attempted or P(last is pears) = P(1st is beans) M2
			Total	11	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
12	a	i	$P(\text{AA or BAA}) = 0.4^2 + 0.6 \times 0.4^2$ oe $= 0.256$ or $\frac{32}{125}$	M1 (AO3.1b) A1 (AO1.1) [2]	allow M1 for either $0.4^2 (\times \dots)$ or $0.6 \times 0.4^2 (\times \dots)$
		ii	ABA or BAB $P(\text{ABA or BAB}) = 0.4^2 \times 0.6 + 0.6^2 \times 0.4$ 0.24 Alternative method $1 - ("0.256" + 0.6^2 + 0.4 \times 0.6^2)$ $= 0.24$	M1 (AO3.1b) M1 (AO1.1) A1 (AO1.1) M1 M1 A1	both seen or implied M1 for either $0.4^2 \times 0.6$ or $0.6^2 \times 0.4$ M1 for $1 - P(\text{A wins or B wins})$ attempted M1 for $1 - ("0.256" + \dots)$ or $1 - ((\dots) + 0.6^2 + 0.4 \times 0.6^2)$ NB $0.4 \times 0.6 = 0.24$: M0M0A0
	b		$'0.256' + '0.24' \times '0.256' + '0.24^2' \times 0.256 + \dots$ $= \frac{0.256}{1-0.24}$ $= \frac{32}{95}$ or 0.337 (3 sf)	M1 (AO3.1b) M1 (AO2.1) A1 (AO1.1) [3]	ft (a)(i)&(ii) ft (a)(i)&(ii) ie $\frac{(a)(i)}{1-(a)(ii)}$ cao $S_5 = 0.337$ SC B1, but with added comment M1M1A1
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance																																																											
13		a	(i) <table><tr><td></td><td></td><td colspan="6">Biased die, Y</td></tr><tr><td></td><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td rowspan="6">Fair die, X</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td><td>11</td><td>13</td></tr><tr><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td><td>14</td></tr><tr><td>3</td><td>5</td><td>7</td><td>9</td><td>11</td><td>13</td><td>15</td></tr><tr><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr></table> (ii) Four 10s circled or otherwise indicated			Biased die, Y								1	2	3	4	5	6	Fair die, X	1	3	5	7	9	11	13	2	4	6	8	10	12	14	3	5	7	9	11	13	15	4	5	6	7	8	9	10	5	6	7	8	9	10	11	6	7	8	9	10	11	12	B2(AO 1.1a 1.1) [2] B1(AO 1.1) [1]	B1 for ≥ 30 values correct Must be exactly four 10s in table.
		Biased die, Y																																																														
		1	2	3	4	5	6																																																									
Fair die, X	1	3	5	7	9	11	13																																																									
	2	4	6	8	10	12	14																																																									
	3	5	7	9	11	13	15																																																									
	4	5	6	7	8	9	10																																																									
	5	6	7	8	9	10	11																																																									
	6	7	8	9	10	11	12																																																									
		b	Outcomes not equally likely oe	E1(AO 2.3) [1]																																																												
		c	$\frac{1}{6} \times 0.14 + \frac{1}{6} \times 0.14 + \frac{1}{6} \times 0.1 + \frac{1}{6} \times 0.01$ $= \frac{13}{200}$ or 0.065	M1(AO 3.1b) A1(AO 1.1) [2]																																																												
		d	$P(S = 10 \text{ \& one score} = 4) =$ $\frac{1}{6} \times 0.14 + \frac{1}{6} \times 0.14 + \frac{1}{6} \times 0.01$ $P(\text{One score} = 4 \mid S = 10)$ $= \frac{P(S=10 \text{ \& one score} = 4)}{P(S=10)} = \frac{29}{600} \div \frac{13}{200}$ $= \frac{29}{39}$ or 0.744 (3 sf)	M1(AO 3.1b) M1(AO 2.1) A1(AO 1.1) [3]	($= \frac{29}{600}$ or 0.0483(3 sf)) ft their (c), dep 1st M1 gained in (d) cao																																																											

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		e	$n = 15$	B1(AO 1.2) [1]	
			Total	10	